Data analyst case study

Retail revenue reconciliation — Kowhai Retail, May 2026

Prepared by ZEN • Tools: Excel, data reconciliation, data visualisation

The challenge

Two reports disagreed on May revenue: the board pack showed about **$1.34M**, an internal store summary showed about **$1.17M**. Leadership needed to know which figure was correct, and why they differed, before a board meeting. I reconciled both against the raw point-of-sale (POS) export of 1,931 transactions.

Tools & techniques used

Tool / technique	How it was used
Excel PivotTables	Aggregated 1,931 transactions to total sales vs refunds, by type and by store.
Data cleaning	Used Remove Duplicates to strip 8 duplicate refund rows (~$1,863) that distorted the total.
Net sales logic	Calculated sales – refunds, correcting a naive total that counted refunds as revenue.
Reconciliation	Compared the raw POS truth against the summary store-by-store to locate the gap.
Root-cause analysis	Isolated the two reasons the summary was short; confirmed the cutoff with the data owner.
Waterfall chart	Visualised how the wrong figure builds up to the correct one in a single view.
Stakeholder reporting	Delivered a one-page memo plus the chart so any reader gets the answer in seconds.

What I did

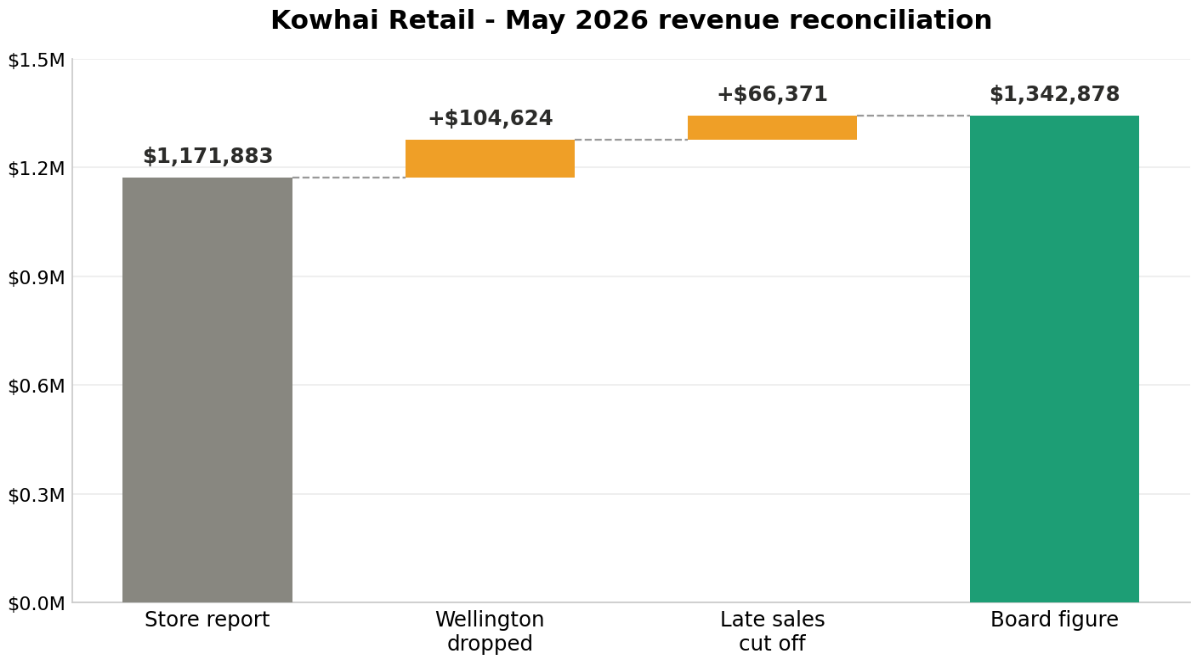
1. **Cleaned the data** — removed duplicate refund rows before any calculation.
2. **Found the true figure** — net sales = sales – refunds = **$1,342,878**, matching the board pack.
3. **Reconciled store-by-store** — nearly every store was slightly short in the summary, pointing to a systematic cause.
4. **Traced root causes** — a renamed store dropped from the report, and an export run before month-end.
5. **Communicated it** — a one-page summary and the waterfall chart below.

What I found

Board pack — correct net sales	**$1,342,878**
Store summary — reported	$1,171,883
Unexplained gap	**$170,995**

• Wellington store dropped (mid-month rename)	$104,624
• Report exported early (last days missing)	$66,371

The result, visualised



Waterfall (bridge) chart: the store report rises to the correct board figure once the dropped store and the missing late sales are added back. Every dollar of the $170,995 gap is accounted for.

Skills demonstrated

Data cleaning • reconciliation • root-cause analysis • data visualisation • communicating findings to non-technical stakeholders.